Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN | Date: 12-01-2026

Time: 02.00Pm to 05.00Pm | GTM-13 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**1**	2	**2**	3	**4**	4	**2**	5	**2**
6	**2**	7	**1**	8	**4**	9	**3**	10	**1**
11	**2**	12	**4**	13	**4**	14	**4**	15	**4**
16	**3**	17	**3**	18	**2**	19	**2**	20	**1**
21	**202**	22	**27**	23	**4**	24	**10**	25	**3**

PHYSICS

26	**4**	27	**3**	28	**2**	29	**3**	30	**1**
31	**4**	32	**3**	33	**2**	34	**3**	35	**1**
36	**4**	37	**2**	38	**4**	39	**3**	40	**4**
41	**3**	42	**1**	43	**1**	44	**4**	45	**1**
46	**3**	47	**25**	48	**7070**	49	**5**	50	**10**

CHEMISTRY

51	**4**	52	**4**	53	**3**	54	**2**	55	**1**
56	**2**	57	**1**	58	**3**	59	**3**	60	**2**
61	**4**	62	**1**	63	**3**	64	**1**	65	**3**
66	**4**	67	**4**	68	**2**	69	**2**	70	**2**
71	**3**	72	**8**	73	**9**	74	**4**	75	**6**

For More Join: @IITJEE_Advanced

SOLUTION
MATHEMATICS

1. Centres : $C_1(1,3)$, $C_2(4,-1)$,

   Radii : $r_1 = r$, $r_2 = \sqrt{16+1-8} = 3$

   $C_1C_2 = \sqrt{9+16} = 5$, $|r_1 - r_2| < C_1C_2 < r_1 + r_2 \Rightarrow |r-3| < 5 < r+3$

   $\therefore r > 2$, $|r-3| < 5 \Rightarrow r-3 < 5 \Rightarrow r < 8$

   $\therefore 2 < r < 8$

2. : If $x = n \in Z$, $|n - 2n| = 4 \Rightarrow n = \pm 4$

   If x= n+K where $0 < K < 1$ then $|n - 2(n+k)| = 4$, , it is possible if $K = \frac{1}{2}$

   $\Rightarrow |-n-1| = 4$

   $\therefore n = 3, -5$

3. Conceptual

4. Since each line of family of lines passes through the focus of given parabola hence shortest intercept is L.R.

5. Conceptual

6. Total selections of four consecutive natural number from first 50 natural numbers

   n(s) = 47

   ∵ S={(1, 2, 3, 6), (2, 3, 4, 5).....(47, 48, 49, 50)}

   Selection of four consecutive natural numbers, if their sum is divisible by 3

   n(E) = 15

   ∵ E={(3n, 3n + 1, 3n + 2, 3n + 3) : n ∈ N}

   E = {(3, 4, 5, 6), (6, 7, 8, 9), (9, 10, 11, 12), ...

   (45, 46, 47, 48)}

   Required probability = $\frac{15}{47}$

7. Expanding along R_1, We get

   $(x+1)\left[x^2 + x\omega + x\omega^2 + \omega^3 - 1\right] - \omega\left[x\omega + \omega^2 - \omega^2\right] + \omega^2\left[\omega - x\omega^2 - \omega^4\right] = 0$

   $\Rightarrow x^3 + x^2\omega + x^2\omega^2 + x^2 = 0$

   $\Rightarrow x^3 + x^2\left(1 + \omega + \omega^2\right) = 0$

   $\Rightarrow x^3 = 0 \quad \Rightarrow x = 0 \quad \left[\because 1 + \omega + \omega^2 = 0\right]$

   Therefore, $x = 0$ is a root of the given equation

8. Median of new set remains the same as that of the original set

9. CONCEPTUAL

10.

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R_1 is not transitive for x=+2, y = 0, z = –2 and

R_2 is not symmetrical as a $\geq$ b does not implies that

b $\geq$ a. Both R_1 and R_2 are not equivalence relations

11.

$\vec{b}-(\hat{a}\cdot\vec{b})\hat{a}=\hat{a}\times(\vec{b}\times\hat{a})$

Now $(\hat{a}\times\vec{b})$ is perpendicular to $\vec{b}-(\hat{a}\cdot\vec{b})\hat{a}=\hat{a}\times(\vec{b}\times\hat{a})$

$\therefore$ one of the angle is 90°.

12. CONCEPTUAL

13. Area bounded by ellipse πab & $2|x|+3|y|\leq 6$ $is 12$ $sq.units$

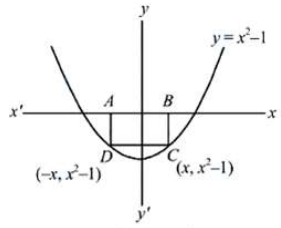
14.

$A=2x\cdot(x^2-1)=2x^3-2x$

$\therefore \quad \frac{dA}{dx}=6x^2-2$

For maximum area $\frac{dA}{dx}=0\Rightarrow x=\pm\frac{1}{\sqrt{3}}$

$\frac{d^2A}{dx^2}=(12x)\Rightarrow\left(\frac{d^2A}{dx^2}\right)_{x=\frac{-1}{\sqrt{3}}}=\frac{-12}{\sqrt{3}}<0$



$\therefore$ Maximum area $=\left|\frac{2}{3\sqrt{3}}-\frac{2}{\sqrt{3}}\right|=\frac{4}{3\sqrt{3}}$

15.

Since, function $f(x)$ is twice differentiable and continuous in $x\in[a,b]$. Then, by LMVT for $x\in[a,c]$

$\frac{f(c)-f(a)}{c-a}=f'(\alpha),\alpha\in(a,c)$

Again by LMVT for $x\in[c,b]$

$\frac{f(b)-f(c)}{b-c}=f'(\beta),\beta\in(c,b)$

$\because \quad f''(x)<0 \Rightarrow f'(x)$ is decreasing

$f'(\alpha)>f'(\beta) \Rightarrow \frac{f(c)-f(a)}{c-a}>\frac{f(b)-f(c)}{b-c}$

$\Rightarrow \frac{f(c)-f(a)}{f(b)-f(c)}>\frac{c-a}{b-c}$ $(\because f(x)$ is increasing$)$

16. ${}^4P_3=24$

17. $\frac{1}{3.6}+\frac{1}{6.9}+\frac{1}{9.12}+.....$

$=\frac{1}{9}\left(\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+-----+\frac{1}{9.10}\right)=\frac{1}{9}\left(1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+---+\frac{1}{9}-\frac{1}{10}\right)=\frac{1}{10}$

18. All Consonants should not be together= Total – All consonants together = 6! – 3! 4! =576

19. We are given A′=A, B′=B

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now, $(AB-BA)'=(AB)'-(BA)'=B'A'-A'B'$

$=BA-AB=-(AB-BA)$ $(AB-BA)'=-(AB-BA)$

Hence, AB - BA is a skew-symmetric matrix.

20. CONCEPTUAL

21.

$f(x)=x^3+x^2f'(1)+xf''(2)+f'''(3),\ x\in R$

$f'(x)=3x^2+2xf'(1)+f''(2),$

$f'''(x)=6$

$\Rightarrow f'''(3)=6$

$f''(2)=6(2)+2f'(1)=12+2f'(1),$

$f'(1)=3(1)^2+2(1)f'(1)+f''(2)=3+2f'(1)+f''(2)$

$\Rightarrow f'(1)=3+2f'(1)+12+2f'(1)$

$\Rightarrow 3f'(1)=-15\Rightarrow f'(1)=-5$

$\therefore f''(2)=12+2(5)=2$

$Now,\ f'(10)=3(10)^2+2(10)f'(1)+f''(2)$

$=300+20(-5)+2=202$

22. So, equation of tangent at $\left(2,\frac{5}{3}\right)$ is $\frac{x(2)}{9}+\frac{y\left(\frac{5}{3}\right)}{5}=1\Rightarrow 2x+3y=9$

Since, this tangent intersect coorditnates axes

$\Rightarrow x-\text{int}\,ercept=9/2,\ y-\text{int}\,ercept=3$

$\therefore\ Area\ \ of\ \ quadrilateral=4\times Area\,of\ \Delta POQ$

$=4\times\frac{1}{2}\times\frac{9}{3}\times 3=27 sq.\ units.$

23. CONCEPTUAL

24. $T_6={}^8C_5\left(x^{-8/3}\right)^3\left(x^2\log_{10}x\right)^5=5600$

$\Rightarrow x^2\left(\log_{10}x\right)^5=100.$

Only x = 10 satisfies this given condition

25. $\frac{dy}{dx}=3x\Rightarrow dy=3xdx$

On integrating, we get $y=\frac{3x^2}{2}+c$

Since, it passes through (-1, -3) then $-3=\frac{3}{2}+c$

$\therefore c=-\frac{9}{2}\qquad \therefore y=\frac{3x^2}{2}-\frac{9}{2}\Rightarrow 2y=3\left(x^2-3\right)$

$\therefore \lambda=3$

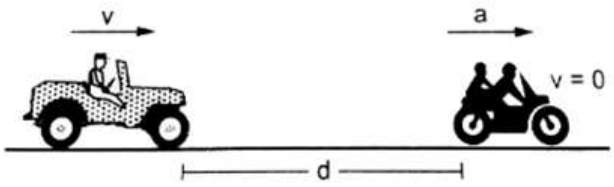
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PHYSICS

26. Energy

27. Paraeelogram Law of force.

28. Suppose the pickpocket is caught at a time t after the motorcycle starts. The distance travelled by the motorcycle during this interval is



$s = \frac{1}{2}at^2 ----i$

During thi s in *terval*

$s + d = vt. ------ii$

By i and ii

$\frac{1}{2}at^2 - vt + d = 0$

$or,\ t = \left(v + \frac{\sqrt{v^2 - 2ad}}{a} \right)$

Thie pickpocket willbe caugth if it real positive.

This will be possible if

$v^2 \geq 2ad \ or\ ,\ v \geq \sqrt{2ad}$

29. R=A/B

30. F-mg = ma
F=m(g+a)
=20000X 15 = 3 X 10^5 N

31. For 4 kg block
4g – T = 4a
For 40 kg block
T – 40g × 0.02 = 40 a
Adding both eq. 40 – 8 = 44a
a=0.7 m/s^2

32. Conceptual

33. Distance covered with 1 step = 1 m
Time taken = 1 s
Time taken to move first 5 m forward = 5s
Time taken to move first 5 m forward = 5s
Time taken to move 3 m backward = 3s
Net distance covered = 5-3 = 2m=
Net time taken to cover 2 m =8s
Net time taken by the drunkard to cover 13 m = 32 + 5 =37 s

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34. Let the speed of the roller coaster at point D be v .
Applying the law of conservation of energy between points A and D ,
we get
$1/2mu^2+mg(2h)$
$=1/2mv^2+0$
or $u^2+4gh=v^2$
or $v=\sqrt{u^2+4gh}$

35. Conceptual

36. Conceptual

37. Conceptual

38. We know that the linear density (μ) is defined as the mass (m) of the wire divided by its length (L):
$\mu=m/L$
From this, we can rearrange the formula to find the length:
$L=m/\mu$
Given:
- Mass of the wire, $m=3.5\times10^{-2}$ kg
- Linear density, $\mu=4\times10^{-2}$ kg/m
Substituting the values:
$L=3.5\times10^{-24}\times10^{-2}=0.875$ m
The fundamental frequency (f) of a vibrating string is given by the formula:
$f=V/2L$
Where V is the speed of the wave on the string. We can also express the speed in terms of tension (T) and linear density (μ):
$V=\sqrt{T/\mu}$
By substituting V into the frequency formula, we have:
$f=1/2L\sqrt{T\mu}$
Squaring both sides gives:
$f^2=1/4L^2\sqrt{T/\mu}$
Rearranging to find T:
$T=4L^2\mu f^2$
Substituting the known values:
- L=0.875 m- μ=4×10−2 kg/m
- f=45 Hz
Calculating:
$T=4\times(0.875)^2\times(4\times10^{-2})\times(45)^2$
Calculating each part:
- $(0.875)^2=0.765625$- $(45)^2=2025$
Putting it all together:
$T=4\times0.765625\times0.04\times2025$ $T\approx248.1$ N

39. The manner in which the two coils are oriented, determines the coefficient of coupling between them. $M = K^2.L_1L_2$ when the two coils are wound on each other, the two coefficient of coupling is maximum and hence mutual inductance between the coils is maximum.

40. Conceptual

For More Join: @IITJEE_Advanced

41. Conceptual

42. Conceptual

43. The intensity of light reflected from upper surface is

$I_0 = I_0 \times 25\ \% = I_0 \times 25/100 = I_0/4$

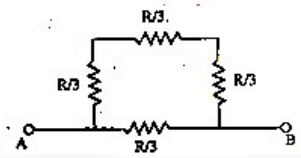
The intensity of transmitted light from upper surface is

$I = I_0 - I_0/4 = 3I_0/4$

$\therefore$ The intensity of reflected light from lower surface is

$I_2 = 3I_0/4 \times 50/100 = 3I_0/8$

$$\therefore \frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2 0} = \frac{\left(\frac{1}{2}+\sqrt{\frac{3}{8}}\right)^2}{\left(\frac{1}{2}-\sqrt{\frac{3}{8}}\right)^2}$$

44. 

R_{net} between AB = $= \frac{R^2}{4R} = 4\Omega$

45. Conceptual

46. Now, find the work done W:

$W = W_2 - W_1 = 4J - 1J = 3J$

47. The speed of light C = $C = \frac{1}{\sqrt{\mu_0 \in_0}} = \frac{1}{\sqrt{2}\times 8} = \frac{1}{4} = 0.25$

48.

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{1}{T}} \text{ or } \frac{N}{10000} = \left(\frac{1}{2}\right)^{\frac{10}{20}}$$

$$\text{or } N = \frac{10000\times 1}{\sqrt{2}} = \frac{10000}{1.414} = 7070$$

49. $W = MB\left[\cos\theta_1 - \cos\theta_2\right]$

Here, $p-n$ junction as forward blased with voltage

50. $= 5 - 3 = 2v$

$\therefore$ *corrent* $I = \frac{2}{200} = \frac{1}{100} = 10^{-2}A = 10mA$

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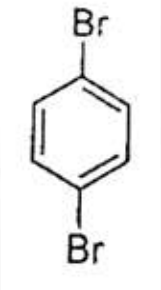
CHEMISTRY

51. he ionic radius in general increases moving top to bottom and further decreases moving left to right. So the correct order is
(Na+,>,Li+,>,Mg^{2+},>,Be^{2+}),(0.98Å,,0.68Å,,0.65Å,)

52. The magnetic moment of $K_3\left[Fe(CN)_6\right]$ is $\sqrt{8}\,B.M$

53. 3>1>2

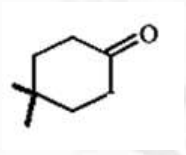
54.



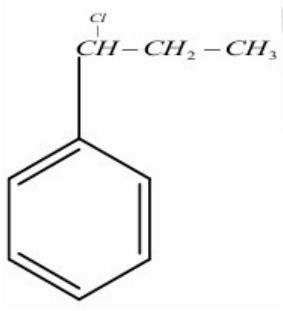
55. $K_c' = \frac{1}{\sqrt{K_c}} = \frac{1}{\sqrt{64}} = \frac{1}{8}$

56. Conceptual

57. Rate of reaction (according to slowest step)
$\Rightarrow r = k_2[\ N_2O_2\ ][\ H_2\ ]$ . . . . (1)
Now for intermediate $[\ N_2O_2\ ]$
$k_1/k-1\ [\ NO\ ]^2$ . . . . (2)
$\Rightarrow [\ N_2O_2\ ] = k_1/k-1\ [\ NO\ ]_2$ . . . . (2)
From equation (1) and (2) $r = k_2 k_1 k_{-1}\ [\ NO\ ]^2\ [\ H_2\ ]$
Overall order of reaction = $k'[NO]^2[H_2]$



58.



For More Join: @IITJEE_Advanced

59. $NH_3 > PH_3 > AsH_3 > SbH_3$-Basic Character

60. In acidic solution [S_2-] is low. So CuS and HgS (IInd group cations) will be precipitated more easily due to low Ksp value.
NiS and MnS (IV group cation) will precipitate if H_2S is passed in basic medium.

61. Chlorobenzene does not undergo hydrolysis on treatment with aqueous NaOH at 298 K.

62. Iodoform test

63.

$$Given: \Delta T_r = 0.558^0 C$$

$$k_r = 1.86 \frac{K \times kg}{mol}$$

$$0.1 m\, aq.sol.$$

$$\Rightarrow \Delta T_f = i \times k_f \times m$$

$$\Rightarrow 0.558 = i \times 1.86 \times 0.1$$

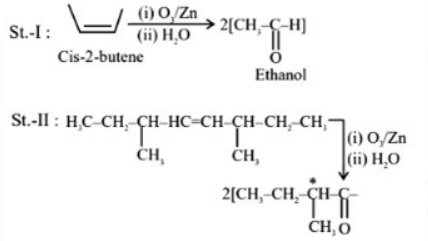
$$\Rightarrow i = 3$$

64. In (a), due to the presence of (4n + 2)πe- it follows Huckel's rule and therefore, it is aromatic.

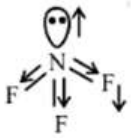
65. 1/2 H_2 + 1/2 $X_2 \longrightarrow HX$
Let the bond enthalpy of X — X bond be x.
$\Delta H_f(HX) = -50$
$1/2\Delta H_{H-H} + 1/2\, \Delta H_{X-X} - \Delta H_{H-X} = 100 kJmol-1$

66. (A) is correct but (R) is not correct.
KCN is ionic and AgCN is covalent.

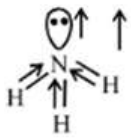
67. Both statement I and Statement II are false



68.



Resultant dipole moment = $0.80 \times 10^{-30} Cm$



Resultant dipole moment = $4.90 \times 10^{-30} Cm$

For More Join: @IITJEE_Advanced

69. Statement I is correct but Statement II incorrect

70. P→iii;Q→iv;R→i;S→ii

71. $E_{cell} = E^0_{cell} - \frac{0.06}{2}\log\left[\frac{Sn^{+2}}{Pb^{-2}}\right]$

72.

$$\text{Molarity} = \frac{(w/w)\% \times \text{density} \times 10}{\text{Molecular wt. of HCl}}$$

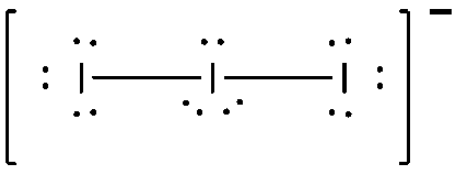
$$= \frac{29.2 \times 1.25 \times 10}{36.5} = 10$$

$M_1V_1 = M_2V_2$

$10 \times V_1 = 0.4 \times 200$

$V_1 = 8\,ml$

73.



74. i , iii, iv, v (optically inactive)

75. $n = 4$,

$m_l = 1 , -1$

Hence l can be" " $= 3 , 2 , 1$

i . e . H_f 2 orbitals

H_d 2 orbitals

Hp 2 Orbitals

Hence total of 6 6 orbitals, and we want

$m_s = -1/2$, that is only kind of spin.

For More Join: @IITJEE_Advanced